

Agentic Text-to-SQL System (FastAPI + PostgreSQL)

Objective

In this assignment, you will learn how to build an **AI-powered agentic system** that:

- Understands natural language questions
- Breaks them into structured components
- Converts them into SQL queries
- Executes queries on a PostgreSQL database
- Handles errors and retries automatically
- Returns meaningful human-readable answers

This simulates a **real-world AI backend system** used in modern applications.

Task 2: Query Understanding (Decomposition Task)

Goal

Learn how to break a natural language question into structured parts before writing SQL.

Instructions

For each question:

1. Read the question carefully
 2. Identify required components:
 - Intent (what is being asked)
 - Tables involved
 - Columns needed
 - Filters/conditions
 - Joins (if any)
 3. Write the decomposition clearly in structured format
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Example

Question:

How many customers are from the USA?

Answer Format:

- Intent: Count total customers
 - Tables: customers
 - Columns: customerNumber
 - Filters: country = 'USA'
 - Joins: None
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Questions to Solve:

- https://docs.google.com/spreadsheets/d/1lgh-Wk6wJMGSEZiQh_ILqVpuRn4UbP_WFUnYzB1RZ9Qc/edit?gid=1012683343#gid=1012683343
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What You Will Learn

- How to think before writing SQL
 - How to break complex queries into simple logic
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